

18 May, 2026

Editor-in-Chief,

Journal of Healthcare Informatics Research

Dear Editor,

Please consider our manuscript entitled “A Systematic Study of Retrieval Pipeline Design for Retrieval-Augmented Medical Question Answering” for publication in the Journal of Healthcare Informatics Research.

This study presents a systematic evaluation of retrieval-augmented generation (RAG) pipelines for medical question answering using the MedQA USMLE benchmark and a structured textbook-based medical corpus. We analyze the impact of retrieval strategies, embedding models, query reformulation, and reranking methods on both accuracy and computational efficiency within a unified experimental framework.

Our results show that retrieval augmentation significantly improves zero-shot medical QA performance, while dense retrieval with query reformulation and reranking achieves the best overall balance between accuracy and efficiency. The work also demonstrates that effective medical RAG systems can be evaluated using modest computational resources.

We believe this manuscript aligns well with the journal’s focus on healthcare AI, medical information retrieval, medical NLP, and AI/ML-based decision support systems.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. AI-assisted tools were used only to improve language clarity and manuscript organization, while all scientific contributions were developed and verified by the authors.

Thank you for your consideration.

Sincerely,

Nusrat Sultana

On behalf of all authors